WORKED SOLUTION

STEP II 2008, Question 3

Pure • Inequalities and Contradiction

STEP II 2008, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q3

The key idea

The sketch shows that the cubic never dips below the x-axis once $x \geq 0$, and rearranging "$y \geq 0$" gives the inequality $x^2(1-x) \leq 4/27$. For the contradiction, notice that the three numbers $bc(1-a)$, $ca(1-b)$, $ab(1-c)$ look awkward one at a time, but their *product* regroups as $a^2(1-a)\cdot b^2(1-b)\cdot c^2(1-c)$, which the first result controls. Part (ii) is the same trick with a quadratic in place of the cubic.

Solution

Part (i): the curve

Differentiating $y = 27x^3 - 27x^2 + 4$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 81x^2 - 54x = 27x(3x-2),$$

which is zero at $x = 0$ and $x = \frac{2}{3}$. At $x = 0$, $y = 4$; at $x = \frac{2}{3}$,

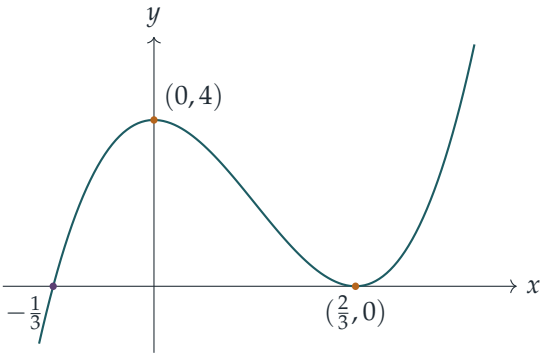
$$y = 27\cdot\tfrac{8}{27} - 27\cdot\tfrac{4}{9} + 4 = 8 - 12 + 4 = 0.$$

Since $\mathrm{d}^2y/\mathrm{d}x^2 = 162x - 54$ is $-54 < 0$ at $x = 0$ and $54 > 0$ at $x = \frac{2}{3}$, the turning points are a maximum at $(0,4)$ and a minimum at $(\frac{2}{3},0)$.

The minimum lies on the x-axis, so the curve touches the axis there and $(3x-2)^2$ is a factor. Comparing the coefficient of x^3 and the constant term gives

$$27x^3 - 27x^2 + 4 = (3x-2)^2(3x+1),$$

so the only other place the curve meets the axis is $x = -\frac{1}{3}$. The curve rises from $-\infty$, crosses the axis at $x = -\frac{1}{3}$, peaks at $(0,4)$, comes down to touch the axis at $(\frac{2}{3},0)$ and then rises again.



Part (i): the inequality

For $x \geq 0$ we have $3x + 1 > 0$ and $(3x - 2)^2 \geq 0$, so the factorisation gives $27x^3 - 27x^2 + 4 \geq 0$. (Equivalently, from the sketch: on $x \geq 0$ the lowest point of the curve is the minimum $(\frac{2}{3}, 0)$.) Rearranging,

$$4 \geq 27x^2 - 27x^3 = 27x^2(1 - x), \qquad \text{so} \qquad x^2(1 - x) \leq \frac{4}{27} \quad \text{for all } x \geq 0,$$

with equality only at $x = \frac{2}{3}$. This is the required result.

Part (i): the contradiction

Suppose, for contradiction, that all three numbers exceed $4/27$:

$$bc(1 - a) > \tfrac{4}{27}, \qquad ca(1 - b) > \tfrac{4}{27}, \qquad ab(1 - c) > \tfrac{4}{27}.$$

All six quantities here are positive, so we may multiply the three inequalities together:

$$bc(1 - a) \cdot ca(1 - b) \cdot ab(1 - c) > \left(\tfrac{4}{27}\right)^3.$$

? **Ask yourself** How does the first result say anything about these mixed products?

Regroup the left-hand side so that each letter is collected with its own bracket:

$$bc(1 - a) \cdot ca(1 - b) \cdot ab(1 - c) = [a^2(1 - a)][b^2(1 - b)][c^2(1 - c)].$$

Each of a, b, c is non-negative, so by the first result each bracket lies between 0 (as $1 - a \geq 0$ and so on) and $4/27$. Multiplying these three inequalities between non-negative numbers,

$$[a^2(1 - a)][b^2(1 - b)][c^2(1 - c)] \leq \left(\tfrac{4}{27}\right)^3.$$

This contradicts the strict inequality above. So our assumption was false, and at least one of $bc(1 - a)$, $ca(1 - b)$, $ab(1 - c)$ is less than or equal to $4/27$.

Part (ii)

The same idea works with a quadratic. For any real t,

$$t(1-t) = \tfrac{1}{4} - (t - \tfrac{1}{2})^2 \leq \tfrac{1}{4},$$

by completing the square. Suppose, for contradiction, that $p(1-q) > \frac{1}{4}$ and $q(1-p) > \frac{1}{4}$. Both sides are positive, so multiplying,

$$p(1-q) \cdot q(1-p) > \tfrac{1}{16}.$$

But the left-hand side regroups as $[p(1-p)][q(1-q)]$, and since $0 < p, q < 1$ each bracket lies between 0 and $\frac{1}{4}$, so the product is at most $\frac{1}{16}$. This is a contradiction, so at least one of $p(1-q)$ and $q(1-p)$ is less than or equal to $\frac{1}{4}$.

Checks

- Equality cases: $a = b = c = \frac{2}{3}$ makes all three numbers in (i) equal to $\frac{4}{9} \cdot \frac{1}{3} = \frac{4}{27}$, and $p = q = \frac{1}{2}$ makes both numbers in (ii) equal to $\frac{1}{4}$. So the bounds $4/27$ and $1/4$ cannot be improved.
- Expanding $(3x-2)^2(3x+1) = (9x^2 - 12x + 4)(3x+1) = 27x^3 - 27x^2 + 4$ confirms the factorisation.
- Numerically, the largest value of $x^2(1-x)$ on $[0,3]$ is $0.148148\ldots = 4/27$. Over a million random triples (a,b,c) in $(0,1)$, the smallest of the three products never exceeded 0.1468, and over a million random pairs the smaller of $p(1-q)$, $q(1-p)$ never exceeded 0.24998.

Common slips

- Multiplying inequalities without saying that everything is positive. The step "$A > \alpha$, $B > \beta$ implies $AB > \alpha\beta$" fails for negative numbers, so the hypothesis $0 < a, b, c < 1$ really is needed.
- Using the bound $x^2(1-x) \leq 4/27$ for $x < 0$, where it is false (try $x = -1$).
- Proving only that one particular number, say $bc(1-a)$, is small. It need not be: with a close to 0 and $b = c$ close to 1 it is close to 1.

Beyond STEP

Taking the product of quantities so that a symmetric bound applies is a close cousin of the AM–GM inequality, which gives the first result directly: for $0 \leq x \leq 1$, applying AM–GM to $\frac{x}{2}, \frac{x}{2}, 1-x$ (which sum to 1) gives $\frac{x^2}{4}(1-x) \leq \left(\frac{1}{3}\right)^3$. The argument also has a pigeonhole flavour: if a product is small, not every factor can be large. These ideas of bounding and optimising are developed in first-year analysis; Abbott's *Understanding*

Analysis is a friendly place to start, and Siklos's *Advanced Problems in Mathematics* has more STEP questions in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.